

AIM:-Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

INPUT:-

```
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import rsa, padding

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

message = input("Enter the message: ").encode()

signature = private_key.sign(
    message,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nDigital Signature Generated Successfully!")

verify_message = input("\nEnter message to verify: ").encode()

try:
    public_key.verify(
        signature,
        verify_message,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
```

```

print("\nSignature Verified Successfully!")
print("Message Integrity: Maintained")
print("Message Authenticity: Verified")

except Exception:
    print("\nSignature Verification Failed!")
    print("Message has been modified or signature is invalid.")

```

OUTPUT:-

```

D:\T079_VIBHUTI > CYBER SECURITY > PRAC4_T079VIBHUTI.py ...
1 from cryptography.hazmat.primitives import hashes
2 from cryptography.hazmat.primitives.asymmetric import rsa, padding
3
4 private_key = rsa.generate_private_key(
5     public_exponent=65537,
6     key_size=2048
7 )
8
9 public_key = private_key.public_key()
10
11 message = input("Enter the message: ").encode()
12
13 signature = private_key.sign(
14     message,
15     padding.PSS(
16         mgf=padding.MGF1(hashes.SHA256()),
17         salt_length=padding.PSS.MAX_LENGTH
18     ),
19     hashes.SHA256()
20 )

```

Signature Verification Failed!
 Message has been modified or signature is invalid.
 PS D:\T079_VIBHUTI\CYBER SECURITY> d:; cd "d:\T079_VIBHUTI\CYBER SECURITY"; & "C:\Users\vmv1u\AppData\Local\Programs\Python\Python314\python.exe" "c:\Users\vmv1u\1\vscode\extensions\ms-python.python-2023.4.1\pythonFiles\lib\python\debugpy\adapter\..\..\debugpy\launcher" "52425" "-..." "D:\T079_VIBHUTI\CYBER SECURITY\PRAC4_T079VIBHUTI.py"

Enter the message:
 PS D:\T079_VIBHUTI\CYBER SECURITY> d:; cd "d:\T079_VIBHUTI\CYBER SECURITY"; & "C:\Users\vmv1u\AppData\Local\Programs\Python\Python314\python.exe" "c:\Users\vmv1u\1\vscode\extensions\ms-python.python-2023.4.1\pythonFiles\lib\python\debugpy\adapter\..\..\debugpy\launcher" "46683" "-..." "D:\T079_VIBHUTI\CYBER SECURITY\PRAC4_T079VIBHUTI.py"

Enter the message: vibhuti

Digital Signature Generated Successfully!

Enter message to verify: vibhuti

Signature Verified Successfully!
 Message Integrity: Maintained
 Message Authenticity: Verified
 PS D:\T079_VIBHUTI\CYBER SECURITY>

GUI**INPUT:-**

```

import tkinter as tk
from tkinter import messagebox
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import rsa, padding

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

```

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```
signature = None

def sign_message():
    global signature

    message = message_entry.get("1.0", tk.END).strip()

    if message == "":
        messagebox.showwarning("Warning", "Please enter a message.")
        return

    signature = private_key.sign(
        message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )

    status_label.config(
        text="Digital Signature Generated Successfully!",
        fg="green"
    )

    signature_text.delete("1.0", tk.END)
    signature_text.insert(tk.END, signature.hex())

def verify_message():
    if signature is None:
        messagebox.showwarning(
            "Warning",
            "Please generate the digital signature first."
        )
        return

    message = verify_entry.get("1.0", tk.END).strip()
```

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```
if message == "":
    messagebox.showwarning(
        "Warning",
        "Please enter a message to verify."
    )
    return

try:
    public_key.verify(
        signature,
        message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )

    result_label.config(
        text="✓ Signature Verified Successfully!\n"
        "✓ Message Integrity Maintained\n"
        "✓ Message Authenticity Verified",
        fg="green"
    )

except Exception:
    result_label.config(
        text="✗ Signature Verification Failed!\n"
        "Message has been modified or signature is invalid.",
        fg="red"
    )

def clear_all():
    message_entry.delete("1.0", tk.END)
    verify_entry.delete("1.0", tk.END)
    signature_text.delete("1.0", tk.END)

    status_label.config(text="")
    result_label.config(text="")
```

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```
root = tk.Tk()
root.title("RSA Digital Signature")
root.geometry("700x700")
root.resizable(False, False)

title = tk.Label(
    root,
    text="RSA DIGITAL SIGNATURE",
    font=("Arial", 20, "bold")
)
title.pack(pady=15)

tk.Label(
    root,
    text="Enter Message:",
    font=("Arial", 12, "bold")
).pack()

message_entry = tk.Text(
    root,
    height=4,
    width=70
)
message_entry.pack(pady=8)

tk.Button(
    root,
    text="Generate Digital Signature",
    command=sign_message,
    font=("Arial", 11, "bold"),
    padx=10,
    pady=5
).pack(pady=5)

status_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
)
```

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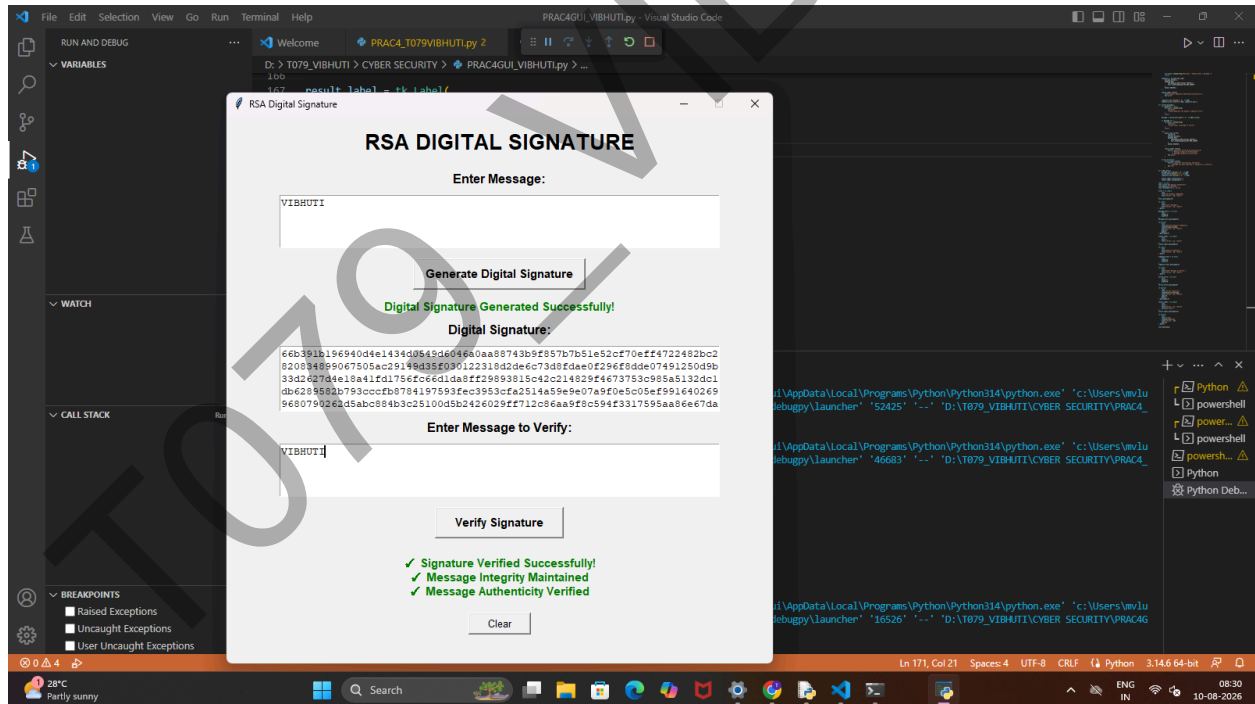
```
)  
status_label.pack(pady=5)  
  
tk.Label(  
    root,  
    text="Digital Signature:",  
    font=("Arial", 12, "bold")  
) .pack()  
  
signature_text = tk.Text(  
    root,  
    height=5,  
    width=70  
)  
signature_text.pack(pady=8)  
  
tk.Label(  
    root,  
    text="Enter Message to Verify:",  
    font=("Arial", 12, "bold")  
) .pack()  
  
verify_entry = tk.Text(  
    root,  
    height=4,  
    width=70  
)  
verify_entry.pack(pady=8)  
  
tk.Button(  
    root,  
    text="Verify Signature",  
    command=verify_message,  
    font=("Arial", 11, "bold"),  
    padx=20,  
    pady=5  
) .pack(pady=5)  
  
result_label = tk.Label(  

```

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```
root,  
text="",  
font=("Arial", 11, "bold"),  
justify="center"  
)  
result_label.pack(pady=15)  
  
tk.Button(  
    root,  
    text="Clear",  
    command=clear_all,  
    font=("Arial", 10),  
    padx=20  
) .pack()  
  
root.mainloop()
```

OUTPUT:-



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